

Structure and Union

- Q. Write a program that uses functions to perform the following operations
- Reading a complex number.
 - Writing a complex number.
 - Addition and sub of 2 complex numbers

```
#include <stdio.h>
```

```
float real;
```

```
float imag;
```

```
} ; struct complex readComplex();
```

```
void writeComplex(struct complex c);
```

```
struct complex addComplex(struct complex c1, struct complex c2);
```

```
struct complex subtractComplex(struct complex c1, struct complex c2);
```

```
int main() {
```

```
    struct complex num1, num2, sum, diff;
```

```
    printf("Enter the first complex number: \n");
```

```
    num1 = readComplex();
```

```
    printf("Enter the second complex number: \n");
```

```
    num2 = readComplex();
```

```
    sum = addComplex(num1, num2);
```

```
    diff = subtractComplex(num1, num2);
```

```
    printf("In second complex number: ");
```

```
    writeComplex(num2);
```

```
    printf("In sum: ");
```

```
    writeComplex(sum);
```

```
printf("In Difference ");
writeComplex(diff);
printf("\n");
return 0;
}
```

```
//
void writeComplex(struct Complex c) {
```

```
    if (c.imag >= 0)
```

```
        printf("%f + %fi", c.real, c.imag);
```

```
    else
```

```
        printf("%f - %fi", c.real, -c.imag); }
```

```
struct Complex addComplex(struct Complex c1, struct Complex c2) {
```

```
    struct Complex result;
```

```
    result.real = c1.real + c2.real;
```

```
    result.imag = c1.imag + c2.imag;
```

```
    return result; }
```

```
struct Complex subtractComplex(struct Complex c1, struct Complex c2) {
```

```
    struct Complex result;
```

```
    result.real = c1.real - c2.real;
```

```
    result.imag = c1.imag - c2.imag;
```

```
    return result;
```

```
}
```

```

exp_6.c > C exp2.c > main()
1  #include <stdio.h>
2
3  struct Employee {
4      char name[50];
5      float basicPay;
6      float da;
7      float grossSalary;
8  };
9
10 int main() {
11     struct Employee emp[100];
12     int i, n;
13
14     printf("Enter the number of employees (up to 100): ");
15     scanf("%d", &n);
16
17     for (i = 0; i < n; i++) {
18         printf("\nEnter details of employee %d:\n", i + 1);
19         printf("Enter name: ");
20         scanf("%s", emp[i].name);
21         printf("Enter basic pay: ");
22         scanf("%f", &emp[i].basicPay);
23
24         emp[i].da = 0.52 * emp[i].basicPay;
25         emp[i].grossSalary = emp[i].basicPay + emp[i].da;
26     }
27
28     printf("\nEmployee Name and Gross Salary Details:\n");
29     printf("-----\n");
30
31     for (i = 0; i < n; i++) {
32         printf("Employee Name: %s\n", emp[i].name);
33         printf("Gross Salary: %.2f\n", emp[i].grossSalary);
34         printf("-----\n");
35     }
36
37     return 0;
38 }
39

```

```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\
output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output> & .\exp2.exe
Enter the number of employees (up to 100): 5

```

```

Enter details of employee 1:
Enter name: sanchita
Enter basic pay: 20000

```

```

Enter details of employee 2:
Enter name: aanya
Enter basic pay: 21390

```

```

Enter details of employee 3:
Enter name: aryan
Enter basic pay: 19028

```

```

Enter details of employee 4:
Enter name: bhumi
Enter basic pay: 30000

```

```

Enter details of employee 5:
Enter name: sehajdeep
Enter basic pay: 18565

```

Employee Name and Gross Salary Details:

```

-----
Employee Name: sanchita
Gross Salary: 30400.00

```

```

-----
Employee Name: aanya
Gross Salary: 32512.80

```

```

-----
Employee Name: aryan
Gross Salary: 28922.56

```

```

-----
Employee Name: bhumi
Gross Salary: 45600.00

```

```

-----
Employee Name: sehajdeep
Gross Salary: 28218.80

```

```

PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>

```

Q. Write a program to compute the monthly pay of 100 employees using each employee's name, basic pay. The DA is computed as 52% of the basic pay. Gross salary (BP + DA). Print the employee's name & gross salary.

```
#include <stdio.h>
struct Employee {
    char name[50];
    float basic pay;
    float gross salary;
};

void read Employee(struct Employee *e);
void compute GrossSalary(struct Employee *e);
void display Employee(struct Employee e);

int main() {
    struct Employee employees[100];
    int n, i;
    printf("Enter the number of employees (max 100);");
    scanf("%d", &n);
    if (n > 100 || n <= 0) {
        printf("Invalid number of employees! \n");
        return 1;
    }
    for (i = 0; i < n; i++) {
        printf("Enter Employee %d \n", i + 1);
        read Employee(&employees[i]);
        compute GrossSalary(&employees[i]);
    }
    printf("\n\n %-5s %-20s %-15s \n", "S.No", "Employee Name", "Gross Salary");
    printf("----- \n");
    for (i = 0; i < n; i++) {
        printf("%-5d %-20s %-15.2f \n", i + 1, employees[i].name, employees[i].gross salary);
    }
    return 0;
}
```


Q. Create a Book structure containing book_id, title, author name and price. Write a C program to pass a structure as a fn argument and print the book details.

```
#include <stdio.h>

struct Book {
    int Book_id;
    char title[100];
    char author[100];
    float price;
};

void displayBook(struct Book b);

int main() {
    struct Book b1;
    printf("Enter book ID: ");
    scanf("%d", &b1.book_id);
    getchar();
    printf("Enter book title: ");
    fgets(b1.title, sizeof(b1.author), stdin);
    printf("Enter book price: ");
    scanf("%f", &b1.price);
    printf("In Book Details: In");
    displayBook(b1);
    return 0;
}

void displayBook(struct Book b) {
    printf("Book ID: %d", b.book_id);
    printf("In Title: %s", b.title);
    printf("Author Name: %s", b.author);
    printf("Price: %.2f\n", b.price);
}
```

```

exp_6.c > C exp4.c > main()
1  #include <stdio.h>
2  #include <string.h>
3
4  union Address {
5      char name[50];
6      char home_address[100];
7      char hostel_address[100];
8      char city[50];
9      char state[50];
10     char zip[10];
11 };
12
13 int main() {
14     union Address a;
15
16     printf("Enter your present address:\n");
17
18     printf("Name: ");
19     scanf("%[^\n]", a.name);
20
21     printf("Home Address: ");
22     scanf("%[^\n]", a.home_address);
23
24     printf("City: ");
25     scanf("%[^\n]", a.city);
26
27     printf("State: ");
28     scanf("%[^\n]", a.state);
29
30     printf("ZIP Code: ");
31     scanf("%[^\n]", a.zip);
32
33     printf("\nYour Present Address is:\n");
34     printf("%s\n", a.name);
35     printf("%s\n", a.home_address);
36     printf("%s, %s - %s\n", a.city, a.state, a.zip);
37
38     return 0;
39 }
40

```

```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output> & .\exp4.exe
Enter your present address:
Name: Sanchita
Home Address: H.NO. 121 New Jain Nagar
City: Yamunanagar
State: Haryana
ZIP Code: 135001

Your Present Address is:
135001
135001
135001, 135001 - 135001
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>

```

Q. Create a union cont 6 strings: name, home-address, hostel-address, city, state and zip. Write a C prog to display your present address

```
#include <stdio.h>
#include <string.h>
char name[50];
char home-address[100];
char hostel-address[100];
char city[50];
char state[50];
char zip[10];
};

int main() {
    union Address info;
    strcpy(info.name, "John Doe");
    printf("Name: %s\n", info.name);
    strcpy(info.hostel-address, "Room No. 25, Sunrise Hostel");
    printf("Hostel Address: %s\n", info.hostel-address);
    strcpy(info.city, "Chennai");
    printf("City: %s\n", info.city);
    strcpy(info.state, "Tamil Nadu");
    printf("State: %s\n", info.state);
    strcpy(info.zip, "600001");
    printf("Zip code: %s\n", info.zip);
    printf("\n**\n");
    printf("Room No. 25, Sunrise Hostel in Chennai, Tamil Nadu-600001\n");
    return 0;
}
```